

ASIF PERVEZ POLOK

📍 Dhaka, Bangladesh

☎ +8801629712967 ✉ asifperveznstu.jsr369@gmail.com 🌐 [Github](#) [LinkedIn](#) [Portfolio](#) [Google Scholar](#)

EDUCATION

Noakhali Science and Technology University

Dec. 2017 – Feb. 2023

Bachelor of Science in Information and Communication Engineering (ICE)

CGPA - 3.26 out of 4.00

Courses - Artificial Intelligence, Digital Image Processing, Calculus and Differential Equation, Statistics for Communication Engineering, Geometry and Vector Analysis, Information Theory

EXPERIENCE

Machine Learning Engineer

Aug 2025 – Present

mPower Social Enterprises Ltd

Dhaka, Bangladesh

- Researching GenAI solutions for mental health in Bangladesh including AI-guided symptom identification and knowledge assistance using fine-tuned LLMs to provide culturally-aligned support for para-counselors, funded by **Wellcome Trust**.
- Developing GenAI and multi-agent systems including **Pandoo** (Voice Pronunciation Evaluation App) using Whisper for language learning, **DB Query & Report Generation** from Natural Language in Bangla/English, and **Bangla Spell Correction System** – enabling advanced language processing and AI-driven automation solutions.

Artificial Intelligence Engineer

May 2025 – July 2025

PetaBytz Technologies Inc

Dhaka, Bangladesh

- Developing GenAI and Agentic AI-based solutions for **Hospital Chatbot** with intelligent, context-aware appointment booking flow and **Multi-Agent Automation System** for developer tasks such as work log creation & insertion, dashboard creation – enabling seamless, end-to-end task execution and conversational AI integration.

Machine Learning Engineer

Dec 2023 – May 2025

Business Automation Ltd

Rajshahi, Bangladesh

- Developed cutting-edge solutions in Generative AI, Computer Vision, and NLP, leveraging advanced neural networks. Contributing to the development of GenAI models on custom data while ensuring seamless deployment and scalability. Proficient in Graph Knowledge & RAG for enhanced AI reasoning, Multi-Agent Systems for autonomous decision-making, Layout-based Segmentation, OCR and LLM Fine-tuning for domain-specific applications.

Artificial Intelligence Training

June 2023 – Sept 2023

Extent IT Ltd

Dhaka, Bangladesh

- Completed 4-month professional AI training program funded by EDGE, IIT, and BCC, focused on algorithm development in ML & DL platform. Led data pre-processing, exploratory data analysis, and feature engineering to boost model performance. Explored NLP and Computer vision core topics and Contributed to projects by applying advanced analytical techniques, extracting actionable insights from complex datasets to drive impactful results.

Research Assistant (Remote)

Nov 2022 – Feb 2023

The University of Texas at El Paso

Texas, United States

- Collaborated with Ph.D. researchers on AI research projects including **Sartorius Cell Instance Segmentation** using UNet with attention mechanisms and **Bangla-BERT Fine-tuning** for news headline classification, contributing to academic publications and advancing computer vision and NLP research initiatives.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

AI Stacks: Machine Learning, Deep Learning, NLP & Computer Vision

Generative AI: Prompt Engineering, LLMs, VLMs, RAG, Vector Database, Knowledge Graph (Neo4j), Langchain, OpenAI, Gemini, Groq, Deepinfra, Ollama, OCR Models

Agentic AI: Langgraph, CrewAI, Phidata

MLOps: MLflow, Git & Github, CI/CD(Jenkins, Github Action), Docker, Prometheus, Grafana, XAI, Whylab & Whylog

Data Analysis & Visualization: Tableau, Pandas, Numpy, Matplotlib, Seaborn, Plotly

Database: MySQL, PostgreSQL, MongoDB

Deployment: Flask, Streamlit, FastAPI

Others: Web Scraping, Time Series Analysis

PROJECTS

NeuroReport: DB Query & Report Generation Multi-Agent System | *Langgraph, LLama3.3, Flask* | [🔗 GitHub](#)

- Developed AI-powered Multi-Agent system that transforms user queries (in both English and Bangla) into PDF report by intelligently understanding user requests, automatically discover and filter relevant database tables, generate optimized SQL queries, execute them against databases, analyze results, and produce PDF reports with charts and insights.

Neo4j-Driven AI Chatbot with Knowledge Graph and LLM Integration | *LLM, Neo4j, HybridSearch* | [🔗 GitHub](#)

- Developed a chatbot system that combines Neo4j Knowledge Graphs, Retrieval-Augmented Generation (RAG), and Hybrid Search to provide contextually accurate responses across PDF, CSV, and TXT datasets. Integrated graph-based reasoning and generative AI for dynamic, real-time querying and scalable conversational experiences, ideal for enterprise applications.

Handwritten Prescription Digitalization using Layout Analysis and OCR | *YoloV8, TrOCR/Florence2, LLM*

- The project focuses on digitizing handwritten prescriptions for medication management through layout analysis and OCR technologies. Employing YOLO model for line-by-line segmentation, it extracts crucial details. A fine-tuned OCR model is utilized for information extraction from handwritten text and LLM model transforms raw data into structured outputs.

DocuFix: Intelligent Document Detection & Rotation Correction | *YoloV11, Flask, OpenCV* | [🔗 GitHub](#) | [🌐 Live](#)

- Developed an intelligent document segmentation and rotation correction solution, utilizing the YOLOv11 segmentation model for document extraction and OpenCV for image processing tasks. Implemented automatic rotation correction to ensure proper alignment, integrated the solution into a Flask web application, and deployed it using Docker.

WhisprNote: Minutes of Meetings from Audio and Video Recording | *Whisper, Gemini-pro* | [🔗 GitHub](#)

- Designed an audio-visual transcription system leveraging Whisper and LLM models to generate clear, actionable minutes of meetings and summarizes meeting transcript. Ensured alignment and accountability by capturing key decisions and action plans in an engaging, accessible format.

BanglaSpellT5: Spell Correction for Bangla Language | *Bangla-T5, HappyTransformer, NLTK* | [🔗 GitHub](#)

- Developed a spell correction system using the Bangla-T5 model by executing comprehensive data preprocessing, model training, and evaluation, resulting in enhanced accuracy and significant improvements in Bangla spell correction tasks.

Automated NID & Visiting Card Cropping System | *YoloV8, OpenCV, Flask* | [🔗 GitHub](#) | [🌐 Live](#)

- Developed a web application using Flask for detecting and cropping visiting and NID cards from uploaded images, integrating OpenCV for image processing and employing a fine-tuned Ultralytics YOLOv8 model with real card and NID images to enhance detection accuracy and seamlessly crop detected regions.

ITSM-Agent: Knowledge-Based Intelligent IT Service Solution | *Langchain, LLM, RAG, Flask* | [🔗 GitHub](#)

- Knowledge-based AI solution leveraging LLM to deliver instant IT support for device setup, password management, VPN configuration, system troubleshooting, software installation, network issues, hardware diagnostics, and many more. Supports multiple concurrent users with maintained session history and context-aware responses.

Name Entity Recognition (NER) with MISTRAL, BERT, and FLAN T5 | *LLMs, Unsloth* | [🔗 GitHub](#)

- Developed Named Entity Recognition (NER) system using MISTRAL, BERT, and FLAN T5 models. The Bangla-Bert model was used to identify and classify entities like person, organizations, locations, objects, and emails, while MISTRAL and FLAN T5 were fine-tuned on a Cyber Security dataset to classify software and security-related entities.

PUBLICATIONS

- Raquib M, Akash MM, Ahmed T, Murad SA, Prity FS, Hossain MA, **Polok AP**, Rahimi N. “A Unified BERT-CNN-BiLSTM Framework for Simultaneous Headline Classification and Sentiment Analysis of Bangla News.” *Under Review at IEEE Transactions on Computational Social Systems*. arXiv:2511.18618. 2025.
- Das NR, Alam M, **Polok AP**, Raquib M, Mamun AA, Hossen MJ. “Social Media Bangla Fake News Detection Using Deep and Machine Learning Algorithms”. *International Journal of Engineering Trends and Technology*, 72(5):346-354. May 2024.

CERTIFICATIONS

- 4-month Professional “Artificial Intelligence Trainee” Program | EDGE, IIT BC | [view](#)
- ICEPC Farewell Contest 2022 organizer award | ICE, NSTU | [view](#)

ACHIEVEMENTS & LEADERSHIP

- Kaggle Notebook Expert | Highest Rank: 1693 | Upvotes: 244 | [k Kaggle](#)
- ICE Sports Club, NSTU | **President** | **Jan 2022 – Feb 2023**